

Jayasri Suresh Vani

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EXECUTIVE SUMMARY

Software Engineer with hands-on industry experience building and deploying AI-powered and data-driven systems across internships and production ready projects. Strong background in backend development, machine learning workflows, and cloud deployments, with practical experience integrating LLMs, optimizing SQL-based data pipelines, and delivering scalable services.

EDUCATION

Purdue University

Master of Science in Computer Science (GPA: 3.89/4.0)

Anna University - Easwari Engineering College

Bachelor of Technology in Information Technology (GPA: 4.0/4.0)

Fort Wayne, Indiana

Aug 2024 - May 2026

Tamil Nadu, India

Sep 2020 - May 2024

WORK EXPERIENCE

AI/ML Research & Development Intern | RAVEN AI | July 2025 - Present | New York, NY

- Advanced medical training evaluation via NLP-driven behavioral scoring across 5 communication dimensions.
- Bench-marked models using GPT-4o, Qwen3, and Llama3, applying prompt engineering and automated loss evaluation (MAE, MSE, R²).
- Engineered 250+ synthetic doctor-patient conversation with patient-name injection and contextual filtering to expand dataset coverage.

Technical Support Engineer | PURDUE UNIVERSITY | Dec 2024 - Present | Fort Wayne, IN

- Orchestrated a campus-wide IP address migration project, transitioning 3000+ devices to a new subnet; facilitated seamless communication and ensured uninterrupted network connectivity for end-users.
- Re-imaged 100+ university systems and deploy software upgrades across 20+ labs, reducing downtime by 25%.
- Provide on-demand technical support, enhancing faculty and student experience in technology-enabled learning spaces.

Data Analyst & Machine Learning Intern | TECHVOLT PVT. LTD | Feb 2023 - Nov 2023 | Coimbatore, India

- Conducted exploratory data analysis (EDA) using Python (Pandas, NumPy, Seaborn) to identify trends, outliers, and patterns, enabling data-driven insights that improved decision-making efficiency by 25%.
- Developed and evaluated diagnostic machine learning models including Logistic Regression, Naive Bayes, and Decision Trees, achieving 85%+ accuracy using Scikit-learn pipelines and cross-validation.
- Applied transfer learning by fine-tuning pre-trained models to improve predictive performance, incorporating data augmentation techniques and advanced visualizations to enhance model robustness and interpretability.

Software Developer Intern | OMEGA BINARIES | Apr 2023 - May 2023 | Chennai, India

- Led a team to build a biometric authentication system via Java Swing, reducing latency by 40% and improving responsiveness by 25%.
- Integrated MySQL with encrypted credential storage and MVC-driven frontend for scalability and usability across 100+ users.
- Integrated MFS100 SDK with the authentication system to provide 100+ user access

PROJECTS

Epileptic Seizure Classification

Jayasri2021/Epileptic-Seizure-Classification

- Developed an end-to-end deep learning pipeline for EEG signals using preprocessing, wavelet transform, CNN/LSTM models, model evaluation and t-SNE representation to detect 6 types of seizure with high accuracy of 92%.

CampusCrew - Centralized Service Platform

Jayasri2021/CampusCrew

- Created a service exchange platform connecting 500+ Purdue students, featuring a React front-end with role-based access and a Flask back-end with real-time Supabase integration. Optimized query latency by 40% and ensured reliability with Cypress/Pytest testing.

Auction-Ease: Online Auction Platform

Jayasri2021/Auction-Ease-Project

- Built and deployed a scalable auction platform supporting 1,000+ bids/min with real-time React interfaces and Java Spring Boot APIs, leveraging ETL pipelines, optimized concurrency control, and load-balanced REST services to reduce response time by 35%.

NewsBias – Bias Detection and Controlled Text Rewriting

Jayasri2021/EAI_News_Media_Bias

- Implemented transformer-based NLP pipeline achieving 92.78% bias detection accuracy along with controlled rewriting using T5-based models with ROUGE-1 score of 0.6053 and explainability using LIME and SHAP.

Adaptive Learning Management System

Jayasri2021/lms_project

- Structured a role-based LMS with secure authentication, automated quiz evaluation, and OLAP-driven analytics for 500+ students; Improved scalability using Docker containers and Prisma ORM for efficient database management.

MoodTunes - <https://moodtunes-423d2.web.app>

Jayasri2021/Music-Playlist-Gneneration-model

- Built a full-stack ML app classifying user emotions with 87% accuracy to generate personalized playlists; deployed on GCP Cloud Run with secure authentication and API integration, and hosted a Logistic Regression model on HuggingFace Space using Gradio SDK.

Gesture2Globe- ALS sign detection

Jayasri2021/Gesture2Globe-ASL-Sign-Detection

- Built an end-to-end ASL recognition system with MediaPipe hand keypoints, TGCN ONNX inference, and a React+Flask pipeline that forms sentences (FLAN-T5), Google translate, and generates bilingual speech via PiperTTS from video and live webcam input.

SKILLS

- **Programming Languages:** Python, Java, TypeScript, JavaScript
- **AI/ML & Data Science:** Deep Learning, NLP, Predictive Modeling, Scikit-learn, TensorFlow, PyTorch, Computer Vision, MediaPipe, Transformers, Sequence Models
- **Frameworks:** React, Next.js, Angular, NestJS, Flask, Django, REST APIs
- **Databases & Cloud:** SQL, PostgreSQL, MongoDB, Firebase, Supabase, Docker, Prisma

CERTIFICATIONS

- **Google Cloud Badge:** *Introduction to Generative AI*
- **Kaggle:** *Into to Machine Learning, Intermediate Machine Learning*
- **Udemy:** *Flutter & Dart, Django4 & Python Fullstack Developer*